

**A Business Data Management Approach For Enhancing
Efficiency And Decision-Making In Kalka Dairy Farm**
A Proposal Report For the BDM Capstone Project

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Declaration Statement

I am Working on a Capstone Project Titled “Analytical Study of a Dairy Farm & Milk Seller”. I Extend my appreciation to **Kalka Dairy Farm** ,for providing the necessary resources that enable me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered from primary sources and Carefully analyzed to assure its reliability.

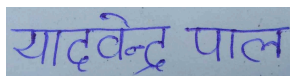
Additionally, I affirm that all procedures employed for the purpose of data collection and collection and analysis have been duly explained in this report. The outcome and derived from the data are an accurate depiction of the findings acquired through analytical procedures.

I am dedicated to adhering to the principles of academic integrity and honesty, and I am receptive to any additional examination or validation of the data contained in this report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary action by relevant authority.

I understand that all recommendations made in this project report are within the context of the academic project taken up towards course fulfilment in the BS Degree program offered by IIT Madras. The Institution does not endorse any of the claims or comments.

Signature of Candidate



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Date : 13/06/2026

1. Executive Summary

Kalka Dairy Farm, Ingohata, Hamirpur, Uttar Pradesh is a small dairy farm with about ten dairy animals. The farm sells fresh milk to customers in the surrounding areas. Like many other small scale dairy farms, the farm operates on a manual register on a day-to-day basis. Data on milk production, animal health, customer payments, expenses and sales are recorded manually. This method has been used for years, but it often leads to a time-consuming record management and increases the chances of errors or missing information

The project is targeted at the development of a simple and organized system for the management of the business data of the farm. There won't be separate registers for different activities, but all the information will be saved in a structured database. These comprise details of dairy animals, daily milk output, customers' details, milk sales and feed purchases, veterinary bills, and other financial transactions. Having all the information together will make it easy to update records, get data, and monitor the farm's performance overall.

Along with data organization, the project focuses on the analysis of the business performance of the farm. To get a sense of the trends in milk production, sales patterns, operating expenses, and overall profitability, a variety of analytical techniques are applied such as descriptive analysis, trend analysis, cost analysis, and performance evaluation. The results are presented in charts, graphs, and dashboards for easy understanding by the farm owner to enable him to make educated decisions about the business.

Microsoft Excel is used for data entry, organization, calculations, and dashboard creation. Python is used to clean, process, analyze, and visualize data.

It is expected that the initiative would offer Kalka Dairy Farm a more effective means of managing its company records and enhance the accuracy of its financial and operational data. But most importantly, it shows that even a small dairy farm can gain from data analysis in finding out where to improve, monitoring business performance and making good decisions for future growth.

2. Organisation Background

Business Name : Kalka Dairy Farm

Address : Ingohata Hamirpur Uttar Pradesh India 210341

Owner Name : Mr. Kalka Prashad

Kalka Dairy Farm is a small dairy farm located in a village called Ingohata, district of Hamirpur, Uttar Pradesh. The farm is established and maintained by Mr. Kalka Prashad. It breeds dairy animals and produces fresh milk to fulfil the daily needs of clients of adjacent villages and local markets. The farm provides quality dairy products and healthy livestock through proper feeding and routine health care measures.

The farm contains about 10 dairy animals and produces an average of 35-40 litres of milk per day with a projected production of 10000 litres per month. Average monthly sales revenue is between ₹60000-₹70000. Depending on seasonal milk yield and market demand, it may fluctuate. The farm also supports the livelihood of the owner and contributes to the local economy by providing a consistent supply of fresh milk and producing employment.

Daily farm operations involve livestock management, milk collecting, feed planning, veterinary care, customer servicing, tracking sales and managing expenses. Currently these operations are recorded manually in registers. This is time consuming and open to error, duplication and difficulties in getting information for business evaluation.

To address these challenges, this project proposes the implementation of a Business Data Management System that will centralize and organize all farm-related information. The system will store records of dairy animals, milk production, customer transactions, feed consumption, healthcare expenses, and financial data in a structured manner. It will also support data analysis through reports and dashboards, enabling the farm owner to monitor operational performance, track income and expenditure, and make informed business decisions. The proposed solution is expected to improve record accuracy, enhance operational efficiency, reduce administrative effort, and provide a strong foundation for the sustainable growth of Kalka Dairy Farm.

3. Problem Statement

3.1 : To establish a centralized business data management system for maintaining accurate and organized records of dairy animals, milk production, customer transactions, feed consumption, and operational expenses.

3.2 : To strengthen operational efficiency and business decision-making by improving financial monitoring, reducing record-keeping errors, and enabling effective analysis of farm performance

4. Background of the problem

Dairy farming is an important source of income for many rural households in India. Small-scale dairy farms, such as Kalka Dairy Farm, depend on effective management of animal records, milk production, sales transactions, feed consumption, and operational expenses to maintain profitability. However, many of these activities are still managed through manual record-keeping methods, including notebooks and paper-based registers.

Manual data management creates several challenges for farm owner Mr. Kalka Prashad. Records may become incomplete, misplaced, or difficult to access when needed. As the volume of information increases, tracking daily milk production, customer payments, animal health records, and business expenses becomes more time-consuming and less accurate. This can affect the quality of business decisions and make it difficult to evaluate the farm's overall performance.

At Kalka Dairy Farm, the absence of a structured data management system may limit the ability to monitor key business activities efficiently. Important information related to production, sales, and expenses may not be readily available for analysis, making it challenging to identify trends, measure profitability, and plan future improvements. In addition, manual calculations increase the risk of errors, which can impact financial records and operational planning.

Therefore, there is a need for a systematic and reliable business data management solution that can organize farm information in a centralized manner. Such a system will help improve record accuracy, simplify data analysis, support informed decision-making, and contribute to the sustainable growth of the dairy farm.

5 Problem Solving Approach

The aim of this project is to develop a systematic Business Data Management System for Kalka Dairy Farm which will assist in effective record keeping and business analysis. The project involves a systematic approach to data collection, data preparation, business analysis, and visualization to identify operational improvements and support in making informed decisions. The approach is structured in three major stages: data collection, analytical methods, and tools for analysis.

5.1. Methods Used with Justification

The following analytical methods will be used to study the business operations of Kalka Dairy Farm.

5.1.1. Descriptive Statistics - Descriptive statistics such as total, average, minimum, maximum, percentage, and frequency will be used for summarizing data.

This method provides a complete picture of milk production, sales, expenses, and customer demand. It helps to know the actual performance of the dairy farm.

5.1.2. Trend Analysis - Trend analysis will look at changes in milk production, sales revenue, expenses, and profit over the three-month study period.

Trend analysis shows business performance that is improving or declining and supports future planning.

5.1.3. Cost Analysis - Cost analysis will be carried out on different types of operational costs like feed cost, veterinary cost, labour charges, electricity bills, transportation cost, and maintenance cost.

Identification of the major cost components and opportunities for cost reduction is enabled by understanding expenditure.

5.1.4. Financial Performance Analysis - We will calculate the revenue, total expenses, gross profit, and monthly profitability.

Financial analysis enables the evaluation of the economic performance of the dairy farm and assists in business planning.

5.1.5. SWOT Analysis

A SWOT Analysis will determine:

- . Strengths
- . Weaknesses
- . Opportunities
- . Threats

SWOT analysis provides strategic recommendations to improve the long-term performance of Kalka Dairy Farm.

5.1.6. Root Cause Analysis

- . Root Cause Analysis will uncover operational problems such as:
- . Bad record keeping
- . Financial records missing
- . Bad inventory tracking
- . Late reporting

Finding the root causes helps us to suggest practical solutions rather than simply treating the symptoms.

5.1.7. Dashboard Analysis - Business dashboards will be summaries of key performance indicators (KPIs).

Dashboards provide the farm owner with a quick visualisation of business performance and assist with decision-making.

5.2. Analysis Tools Used with Justification

The project utilizes the following software tools for data management and analysis:

Microsoft Excel

- . Data entry and organization

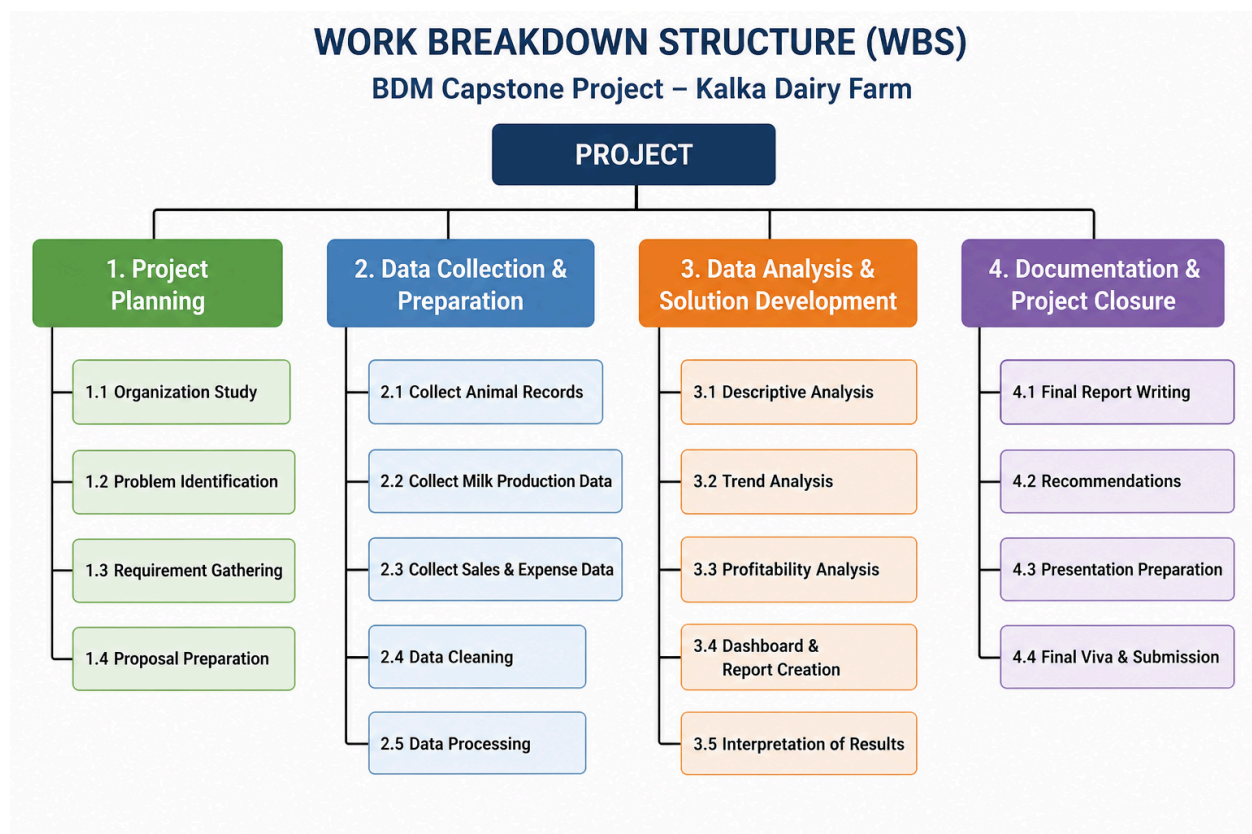
- . Data cleaning and validation
- . Formula-based calculations
- . Pivot Tables
- . Charts, graphs, and dashboards
- . Financial summaries and reporting

Python

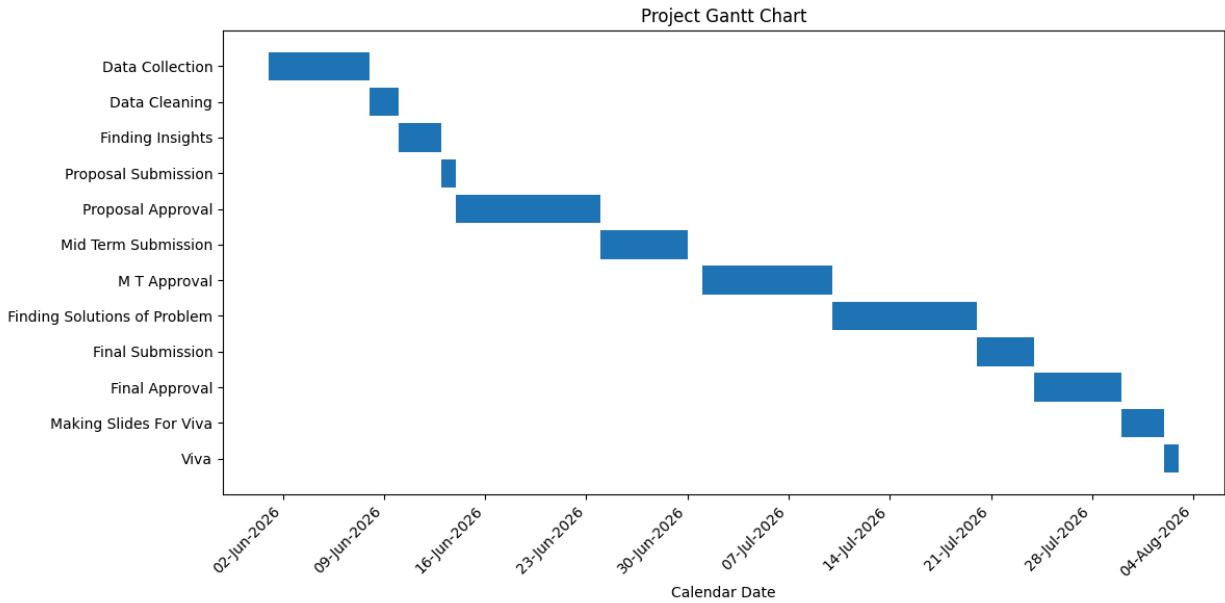
- . Data preprocessing and cleaning
- . Statistical analysis
- . Automated report generation
- . Data visualization using libraries such as Pandas and Matplotlib

The combination of Microsoft Excel and Python provides an efficient platform to organize, analyse, and visualize business data. Such tools help Kalka Dairy Farm maintain accurate records, analyse data meaningfully and generate reports to enhance operational and financial performance.

6. Expected Timeline



(fig1)



Gantt Chart(Fig 2)

7. Expected Outcome

1. Improved Data Management and Accuracy

The proposed system is expected to organize farm records in a centralized manner, improving data accuracy and reducing errors associated with manual record-keeping.

2. Enhanced Business Analysis and Decision-Making

The project will provide meaningful insights into milk production, sales, and operational expenses through data analysis and reporting, supporting informed business decisions.

3. Increased Operational Efficiency and Business Growth

The system is expected to streamline record management processes, save time, improve financial monitoring, and contribute to the sustainable growth of Kalka Dairy Farm.